

Prabin Kumar Rath

+1 602-884-4471 • prath4@asu.edu • [Portfolio](#) • [LinkedIn](#) • [Github](#) • [Scholar](#)
Tempe, AZ, 85281

EDUCATION

Ph.D. Computer Science

Arizona State University, Tempe, AZ

Aug 2025 - May 2030

4.00 GPA

M.S. Artificial Intelligence and Robotics

Arizona State University, Tempe, AZ

May 2024

4.00 GPA

B.S. Computer Science

National Institute of Technology, Rourkela, India

May 2020

8.91 GPA

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C/C++, C#, TypeScript, Java, Shell

Cloud Technologies: Azure, AWS, Kubernetes, Docker **Simulators:** IsaacLab, PyBullet, CoppeliaSim, MuJoCo, Maniskill

Tools and Frameworks: PyTorch, TensorFlow, ROS/ROS2, OpenCV, Open3D, Git, Linux, Claude Code

PUBLICATIONS

- [1] "Self-Supervised Keyframe Discovery for Horizon-Invariant Behavior Cloning", In *NeurIPS 2026*
- [2] "StageCraft: Execution Aware Mitigation of Distractor and Obstruction Failures in VLA Models", In *IROS 2026* [↗](#)
- [3] "Factorizing Diffusion Policies for Observation Modality Prioritization", In *ICRA 2026* [↗](#)
- [4] "XMoP: Whole-Body Control Policy for Zero-shot Cross-Embodiment Neural Motion Planning", In *ICRA 2025* [↗](#)

RESEARCH EXPERIENCE

Vision-Language-Action (VLA) Models for Franka FR3 Robot | Systems Engineering

Fall 2025 - Present

- Extended the LeRobot backend to support the Franka FR3 robot platform through ROS2 APIs.
- Created a real-time teleoperation system with multiple control interfaces, including Meta Quest 3, GELLO whole-body leader-follower setup, and SpaceMouse.
- Developed robust and fault-tolerant robot teleoperation software for HRI user study experiments.
- Trained, deployed, and benchmarked multiple VLA and behavior cloning models, including $\pi_{0.5}$, SmolVLA, XVLA, Groot N1, DP, and ACT on the Franka FR3 robot, achieving 90%+ success rates on long-horizon manipulation tasks.

Policy Memory for Horizon-Invariant Behavior Cloning | NeurIPS 2026

Fall 2025 - Spring 2026

- Proposed a novel self-supervised algorithm for memory-intensive behavior cloning that discovers sparse, decision-relevant keyframes in non-Markovian expert demonstrations without explicit keyframe annotations.
- Demonstrated behavior cloning with 100% SR at 1M-step horizons, outperforming SOTA baselines by 13.9% absolute SR on MIKASA-Robo and retaining 80% SR at 20 $\times$ longer horizons on a real robot. [🌐 Project](#)

Post-training Deployment Robustness under Visual Distribution Shifts | ICRA, IROS 2026

Fall 2024 - Spring 2026

- Investigated prompting strategies for strict instruction following, spatial environment understanding, and low-hallucination failure reasoning in vision-language models (VLMs).
- Developed a training-free VLA improvement method leveraging VLM reasoning to anticipate and prevent failures, boosting success rates by 23% for $\pi_{0.5}$ and 54% for SmolVLA under visual disruptions and obstructions. [🌐 Project](#)
- Improved Diffusion Policy robustness to visual distractors, camera occlusions, and appearance changes through sensor modality prioritization, outperforming baselines by 15% in simulation and 40% on real-world tasks. [🌐 Project](#)

Cross-Embodiment Generalization in Neural Motion Planning | ICRA 2025

Fall 2023 - Spring 2024

- Designed a novel behavior cloning policy architecture for neural motion planning, enabling zero-shot configuration-space generalization across 7 commercial robots with a 70% average planning success rate.
- Introduced a pointcloud-based semantic segmentation method for detecting collision points on robot surfaces, achieving cross-embodiment sim-to-real generalization with 76.6% IoU and 90.3% recall across 7 robots.
- Demonstrated sim-to-real motion planning from pointclouds on Franka FR3 and Rethink Sawyer robots, achieving a 72% average planning success rate in cluttered environments. [🌐 Project](#)

INDUSTRY EXPERIENCE

Experian, Costa Mesa, CA:

MLOps Engineer

June 2024 - May 2025

- Developed scalable workflows for end-to-end ML model lifecycle management and built inference orchestration platforms meeting millisecond-latency response SLAs while serving 300+ enterprise clients. [🌐 Product](#)

Wells Fargo, Hyderabad, India:

Software Engineer

Aug. 2020 - July 2022

- Architected and deployed robust, multithreaded REST API modules in Python, enabling efficient integration with Azure Cognitive Services. Optimized average response time by 75%, while delivering 4x throughput.

SERVICE

Paper Reviewer: CoRL 26 | ICRA 26, 25, 24 | RA-L 26, 23 | IROS 26, 23 | RO-MAN 23

Session Chair: Autonomous Vehicles @ ICRA 2025